

Unit 2, Lesson 4: Rewriting Sums Using the Distributive Property

Benchmark

MA.6.NSO.3.2 Rewrite the sum of two composite whole numbers having a common factor, as a common factor multiplied by the sum of two whole numbers.

Learning Target

- ☐ I can rewrite the sum of two composite numbers with a common factor using the distributive property.

2.4.1 Warm-Up: Secret Number

Juanita has a secret number. Read her clues and answer the questions that follow.

Clue 1: My secret number is a factor of 60.

1. What is the smallest value Juanita's number could be?
2. What is the largest value Juanita's number could be?

Clue 2: My number is prime.

3. What values are possible for Juanita's number at this point?

Clue 3: 15 is a multiple of my secret number.

Clue 4: My secret number is a factor of 20.

4. What is Juanita's secret number?
5. Could you have determined her secret number only using the first three clues? Explain.

2.4.2 Collaboration: Equivalent Numeric Expressions

In each equation, two numeric expressions are set equal to each other.

1. With your partner, discuss whether each equation is true or false and complete the table.

Equation	Is it true or false?
$42 + 12 = 6(7 + 2)$	
$6(7 + 2) = (6 \cdot 7) + (6 \cdot 2)$	

2. How is 6 related to 42 and 12?
3. How do the values 7 and 2 relate to the original values 42 and 12?

2.4.3 Guided Instruction: Rewriting Sums Using Distributive Property

Sums of **composite whole numbers** can be rewritten using the distributive property and a common factor of the numbers.



A **composite number** is a whole number greater than 1 that has at least one whole-number factor other than one and itself.

1. What are the common factors of 16 and 24?

Factors of 16	
Factors of 24	
Common Factors of 16 and 24	

2. Isaiah rewrote the sum $16 + 24$ using the distributive property. Each step in his thinking is shown. Complete the descriptions of Isaiah’s process by examining his thinking and work. Use the words in the bank below to complete each description.

distributive equivalent factor multiplication

Isaiah’s Thinking (💭) and Work (✍️)	Description
 “I know 2 is a common factor of both 16 and 24.”	Isaiah identified a _____ both numbers have in common.
 $16 + 24 = 2(8) + 2(12)$	He rewrote the expression using _____ with the common factor.

 $2(8) + 2(12) = 2(8 + 12)$	He rewrote the expression using the _____ property.
 "The sum $16 + 24$ can be rewritten as $2(8 + 12)$ using the distributive property."	Isaiah recognized that $16 + 24$ and $2(8 + 12)$ are _____ numeric expressions.

3. Rewrite the sum $16 + 24$ using the distributive property in two additional ways.

 $16 + 24 = \underline{\hspace{1cm}} (\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$

 $16 + 24 = \underline{\hspace{1cm}} (\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$
4. Two students made errors rewriting the sum $54 + 36$ using the distributive property. Each student’s expression is shown.

Rewrite $54 + 36$ using the distributive property.	
Naveah’s Expression	$4(14 + 9)$
Gavin’s Expression	$3(18 + 36)$

Work with your partner to answer the following questions.

- a. What error did Naveah make?
- b. What error did Gavin make?

- c. Generate two different equivalent expressions for the sum $54 + 36$ by rewriting it using the distributive property.
5. Rewrite an equivalent expression for the sum of $40 + 64$ using the distributive property with the **greatest common factor** (GCF).

2.4.4 Collaboration: Missing Values

1. Work with your partner to determine the missing values that make each equation true.

a. $40 + 16 = 4(10 + \underline{\hspace{1cm}})$

b. $15 + 80 = 5(\underline{\hspace{1cm}} + 16)$

c. $108 + 198 = \underline{\hspace{1cm}}(6 + 11)$

d. $28 + 63 = 7(\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$

e. $45 + 18 = \underline{\hspace{1cm}}(\underline{\hspace{1cm}} + 6)$

f. $135 + 27 = 9(\underline{\hspace{1cm}} + 3)$

2. Choose one of the equations from question 1 and explain how you determined the missing value(s).

To find the missing value in equation , I

_____.

3. Have your partner read your explanation and offer suggestions on how to improve it.

2.4.5 Your Turn!

1. Rewrite an equivalent expression for each sum using the distributive property with a common factor.

a. $51 + 21$

b. $124 + 20$

c. $72 + 24$

d. $39 + 65$

2. Rewrite an equivalent expression for the sum of $68 + 34$ using the distributive property with the **greatest common factor** (GCF).

2.4.6 Wrap-Up: Ticket Out the Door

1. Rewrite an equivalent expression for the sum of $75 + 30$ in two different ways using the distributive property and a common factor.
2. Complete the equation with the missing values that make the equation true.

$$49 + 35 = \underline{\hspace{1cm}} (7 + \underline{\hspace{1cm}})$$

